Project Report
Band-Pass Filter Frequency Response Analysis

Daniel Shaheen
Spring 2026

***Abstract*—The objective of this final project was to take an unknown frequency response and design a replacement band-pass circuit. This involved taking measured data, estimating a transfer function based off of that data, building a circuit model in LTspice, and comparing the measured data, estimated transfer function, and simulated circuit data.**

***Index Terms*—band-pass filter, frequency response, MATLAB, LTspice, transfer function**

I. Introduction

The objective of this final project was to take an unknown frequency response and design a replacement band-pass circuit. This involved taking measured data, estimating a transfer function based off of that data, building a circuit model in LTspice, and comparing the measured data, estimated transfer function, and simulated circuit data.

The measured data was given as MATLAB data files. MATLAB was used to graph the measured response, convert magnitude values to decibels, locate the peak gain, find the -3dB cutoff frequencies, and create an estimated transfer function. LTspice was then used to design a circuit that would behave approximately the same way, and simulate that circuit.

This project bridges the gap between network analysis theory and realistic circuit design. A transfer function allows us to describe what we mathematically expect the response of a system to be. However, simulating that system as an actual circuit in LTspice allows us to see how it behaves when real world limitations of op-amps and actual component values are added. By comparing results from MATLAB and LTspice we can determine if our replacement circuit is "good enough" compared to the original measured response.

II. Procedure

The project was completed using the following steps:

- Loaded the provided measured data into MATLAB.
- Converted the measured magnitude into decibels using $20\log_{10}(M)$.
- Plotted the measured magnitude response.
- Found the peak gain and the -3 dB cutoff frequencies.
- Estimated a transfer function based on the measured response.
- Designed the LTspice circuit using standard resistor and capacitor values.
- Ran an AC sweep simulation in LTspice using `.ac dec 200 1 1G`.
- Plotted $V(out)/V(in)$ magnitude and phase in LTspice.
- Used the LTspice cursor tool to verify the passband gain.
- Exported the LTspice magnitude and phase data.
- Imported the LTspice output into MATLAB.
- Compared the measured, estimated, and LTspice magnitude responses.
- Compared the estimated and LTspice phase responses in MATLAB.

III. Measured Data Analysis

The measured magnitude was converted to decibels using:

$$M_{dB} = 20\log_{10}(M)$$

where M is the measured magnitude in V/V.

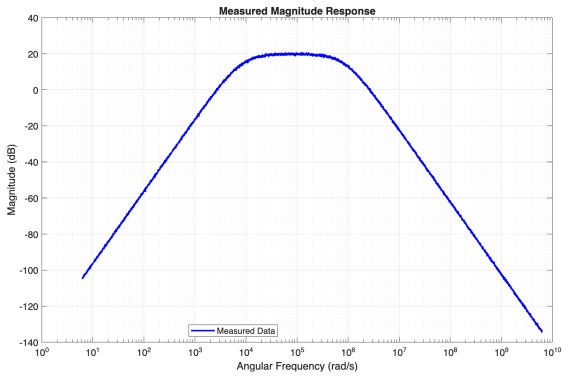


Fig. 1. Measured magnitude response from the provided data.

The measured response shows a clear band-pass filter shape. Low frequencies are attenuated, midband frequencies have high gain, and high frequencies are attenuated again.

IV. Calculations

The peak gain was found from the measured magnitude response:

$$M_{peak,dB} = \max(M_{dB})$$

From the MATLAB result, the peak gain was:

$$M_{peak,dB} = 20.543 \text{ dB}$$

The voltage gain corresponding to this value is:

$$K = 10^{\frac{20.543}{20}} = 10.645$$

For the practical circuit design, the gain was rounded to:

$$K \approx 10$$

The -3 dB cutoff level was calculated as:

$$M_{cutoff} = M_{peak,dB} - 3$$

$$M_{cutoff} = 20.543 - 3 = 17.543 \text{ dB}$$

The lower and upper cutoff frequencies were found where the measured response crossed this -3 dB level.

$$\omega_{cL} = 1.3175 \times 10^4 \text{ rad/s}$$

$$\omega_{cH} = 4.5952 \times 10^5 \text{ rad/s}$$

The approximate slopes were:

$$\text{Low-frequency slope} \approx 39.66 \text{ dB/dec}$$

$$\text{High-frequency slope} \approx -39.72 \text{ dB/dec}$$

These slopes suggest two high-pass poles and two low-pass poles, so the circuit was modeled as a second-order high-pass section cascaded with a second-order low-pass section.

TABLE I
SUMMARY OF MAIN MATLAB-CALCULATED PARAMETERS.

Parameter	Value
Peak gain	20.543 dB
Peak gain	10.645 V/V
Peak frequency	1.2111×10^5 rad/s
-3 dB level	17.543 dB
Lower cutoff frequency	1.3175×10^4 rad/s
Upper cutoff frequency	4.5952×10^5 rad/s
Low-frequency slope	39.66 dB/dec
High-frequency slope	-39.72 dB/dec
Chosen design gain	$K = 10$ V/V
RMS magnitude error	2.371 dB

The corrected estimated corner frequencies used in the transfer function were:

$$\omega_1 = 8479.39 \text{ rad/s}$$

$$\omega_2 = 713988.87 \text{ rad/s}$$

The estimated transfer function was modeled as:

$$H(s) = K \left(\frac{s}{s+\omega_1} \right)^2 \left(\frac{\omega_2}{s+\omega_2} \right)^2$$

Substituting the estimated values:

$$H(s) = 10 \left(\frac{s}{s+8479.39} \right)^2 \left(\frac{713988.87}{s+713988.87} \right)^2$$

The op-amp gain stage was designed using the non-inverting gain equation:

$$A_v = 1 + \frac{R_f}{R_g}$$

Using:

$$R_f = 8.2k\Omega$$

$$R_g = 910\Omega$$

$$A_v = 1 + \frac{8200}{910}$$

$$A_v \approx 10.01$$

In decibels:

$$A_{v,dB} = 20 \log_{10}(10.01)$$

$$A_{v,dB} \approx 20.0 \text{ dB}$$

This matches the desired passband gain.

The LTspice cursor also verified the gain. From the LTspice plot, the cursor showed approximately:

$$f \approx 24.5 \text{ kHz}$$

$$M \approx 19.69 \text{ dB}$$

This value is close to the expected 20 dB gain.

V. PEAK AND CUTOFF FREQUENCIES

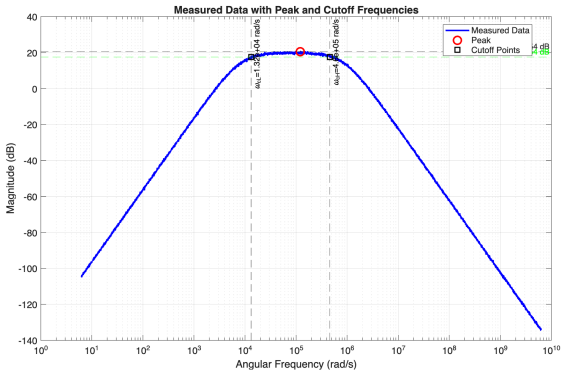


Fig. 2. Measured response with peak and -3 dB cutoff frequencies marked.

The marked points show the peak gain and the lower and upper -3 dB cutoff frequencies. These two cutoff frequencies define the passband of the filter.

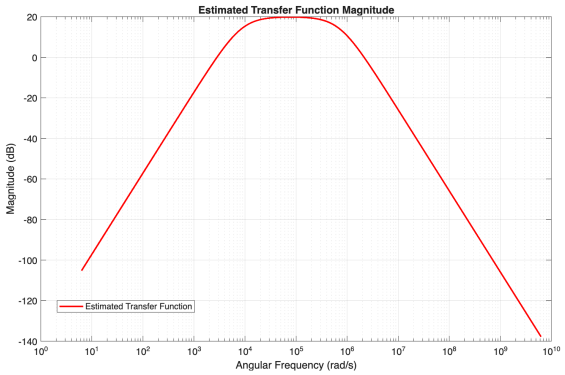


Fig. 3. Estimated transfer function magnitude response.

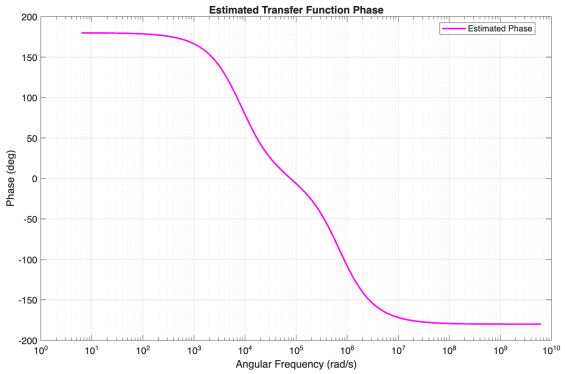


Fig. 4. Estimated transfer function phase response.

VI. Estimated Transfer Function Results

The estimated magnitude response follows the expected band-pass shape. It rises at low frequencies, stays near 20 dB in the passband, and rolls off at high frequencies.

The estimated phase starts near positive phase, transitions through the passband, and approaches negative phase at high frequency. This is expected for a band-pass transfer function.

VII. LTspice Circuit and Simulation

The LTspice circuit used cascaded high-pass stages, a gain stage, and low-pass stages.

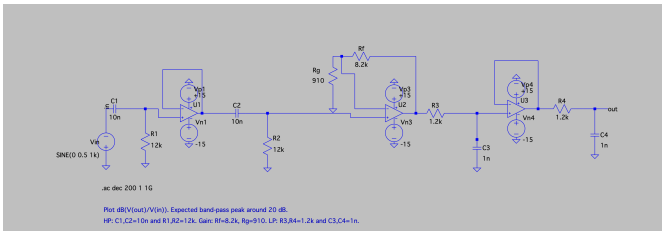


Fig. 5. LTspice band-pass filter circuit.

The signal first enters the high-pass section of the circuit. The first high-pass stage consists of C_1 and R_1. The second high-pass stage consists of C_2 and R_2. The job of these two stages is to attenuate low-frequency signals and pass higher-frequency signals. Having two high-pass sections gives a low-frequency slope of roughly $+40$ dB/dec, which agrees with what was found from our MATLAB data.

After passing through the high-pass sections, the signal then enters the op-amp gain stage. This gain stage uses $R_f = 8.2k\Omega$ and $R_g = 910\Omega$. This is a non-inverting amplifier stage, so its gain can be calculated using:

$$A_v = 1 + \frac{R_f}{R_g}$$

This produces a gain of around 10.01 V/V, or roughly 20 dB. This value controls the level of the passband for this circuit.

The last stage of the circuit is the low-pass section. The first low-pass stage uses R_3 and C_3. The second low-pass stage uses R_4 and C_4. The low-pass stages attenuate higher-frequency signals. Having two low-pass sections gives a high-frequency slope of roughly -40 dB/dec which agrees with our measured response. Cascading the two high-pass stages, gain stage, and two low-pass stages results in the desired active band-pass response.

The command used for the LTspice AC sweep was:

```
.ac dec 200 1 1G
```

The transfer function simulated was:

$$\frac{V(out)}{V(in)}$$

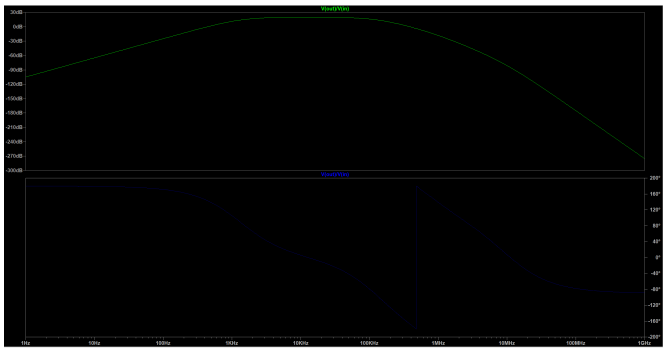


Fig. 6. LTspice magnitude and phase response for $V(out)/V(in)$.

Examining the LTspice magnitude plot we can see the characteristic shape of a band-pass filter. As frequency increases, the response increases until it reaches the filter's passband and plateaus at a maximum of about 19.69 dB. As frequency increases further, the response begins to decrease. The frequency at which this begins was measured using the LTspice cursor tool to be around 24.5 kHz, and we designed our circuit to have a gain of 20 dB so this is close enough. We can also note how the phase of our circuit shifts as a function of frequency. At low frequencies, the circuit's phase is positive but then becomes negative at high frequencies. The reason for the jump up or down in the vertical direction of the LTspice

plot is that the phase wraps from $+180^\circ$ to -180°. This is normal and is not because there is an error in our circuit.

VIII. Magnitude Response Comparison

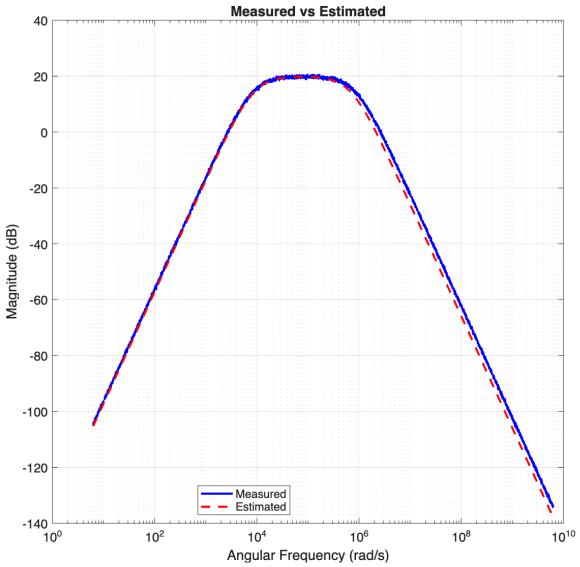


Fig. 7. Measured vs estimated magnitude response.

The measured and estimated magnitude responses match closely. This shows that the estimated transfer function is a good approximation of the measured data.

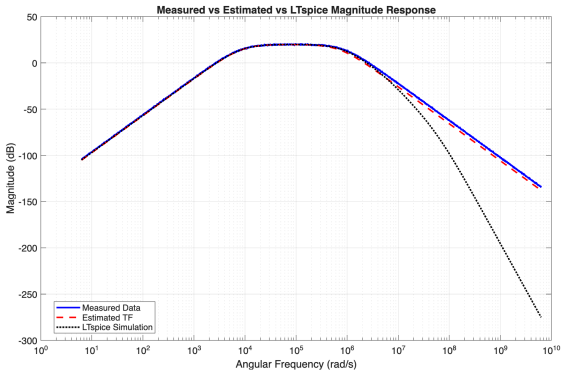


Fig. 8. Measured, estimated, and LTspice magnitude response comparison.

The measured, estimated, and LTspice responses agree well in the passband. At very high frequencies, LTspice rolls off faster. This is expected because the simulated op-amp has non-ideal high-frequency behavior.

IX. Phase Response Comparison

The phase response was plotted in MATLAB using degrees on the vertical axis and angular frequency in rad/s on a logarithmic horizontal axis. The estimated transfer function phase and LTspice phase were plotted on the same figure with different line styles and a legend.

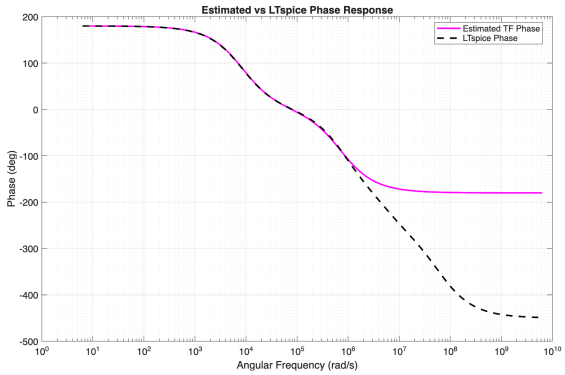


Fig. 9. Estimated transfer function and LTspice circuit simulation phase responses.

At low and mid frequencies, the estimated phase and LTspice phase match well. At high frequencies, the LTspice phase shifts more than the estimated transfer function. This difference is reasonable because LTspice includes non-ideal op-amp bandwidth and extra high-frequency poles, while the estimated transfer function is a simplified model.

X. Discussion

The primary discovery of this project is that the measured system is a band-pass filter. As shown in Fig. 1, the measured magnitude plot rose at low frequency, had a flat passband around 20 dB, then fell at high frequency. Section 4 includes a table of parameters which shows the important values found from MATLAB. The peak gain was 20.543 dB. The two cutoff frequencies were near 1.3175×10^4 rad/s and 4.5952×10^5 rad/s. The slopes were near $+40$ dB/dec and -40 dB/dec which agrees with using two high-pass stages and two low-pass stages in the replacement circuit. The biggest challenge with this project was exporting data from LTspice and importing it into MATLAB. LTspice gives magnitude and phase as one column of text, so the exported data had to be cleaned up and separated in MATLAB before it could be plotted. Another interesting thing I noticed was the jump in the LTspice phase plot. It was not due to a problem in the circuit. Phase wrapping happens when phase values shift suddenly between $+180^\circ$ and -180°, and this is what caused the jump. Our measured magnitude plot lines up well with the estimated magnitude plot, so our transfer function estimate was reasonable. The LTspice simulation agreed with the passband as well. The largest difference was at high frequency where LTspice rolled off faster with more phase shift. This makes sense because the LTspice model of the op-amp has bandwidth limitations and extra poles. For this project, I learned how to go from measured frequency response data to an estimated transfer function. I also learned how to take a transfer function and create a circuit to match that transfer function. I also discovered that MATLAB and LTspice won't always match

up because LTspice includes more realistic circuit effects. My recommendation for the next person would be to build the replacement part using the LTspice circuit provided in this report. This circuit works because it uses common resistor and capacitor values, has about 20 dB of gain in the passband, and matches the desired band-pass shape. If the replacement part needed to more accurately match at very high frequencies, I would use a higher bandwidth op-amp or tweak the low-pass section after testing.

XI. Conclusion

A band-pass filter was examined through measured data, estimated transfer function, and LTspice data. The peak gain, cutoff frequencies, and transfer function behavior were analyzed with MATLAB. LTspice was utilized to design and simulate the circuit. Measured and estimated response aligned very well. LTspice also aligned with the main band-pass behavior. There was extra roll-off of high frequencies due to the limitations of real op-amps. The circuit and MATLAB correctly modeled the replacement band-pass network.

XII. Feedback

I learned how MATLAB and LTspice could be used together to analyze a circuit. MATLAB was useful for processing measured data, determining cutoff frequencies, and comparing responses. LTspice was useful for verifying that the circuit was designed to have the desired band-pass response. The part I had trouble with was importing the LTspice data into MATLAB. The magnitude and phase were exported together so it took some doing to separate the two. An example that shows how to separate LTspice data would help this project be easier to follow.

APPENDIX A: MATLAB CODE

```
% Put this file in the same Project 1 folder as:
%   datafiles/data_x.mat
%   datafiles/data_y.mat
%   LTSpice_Output.txt

clear; clc; close all;

%% Task 2: Load Measured .MAT Data
Sx = load('datafiles/data_x.mat');
Sy = load('datafiles/data_y.mat');

w_meas = Sx.dx_omega(:);      % rad/s
mag_meas = Sy.dyn(:);         % V/V
mag_meas_dB = 20*log10(mag_meas);

%% Automatic Measurements
[peak_dB, idx_peak] = max(mag_meas_dB);
peak_mag_vv = mag_meas(idx_peak);
w_peak = w_meas(idx_peak);

cutoff_dB = peak_dB - 3;

idx_low = find(mag_meas_dB(1:idx_peak) >= cutoff_dB, 1, 'first');
w_low_cut = interp1(mag_meas_dB(idx_low-1:idx_low), ...
                    w_meas(idx_low-1:idx_low), cutoff_dB);

idx_high_local = find(mag_meas_dB(idx_peak:end) <= cutoff_dB, 1, 'first');
idx_high = idx_peak + idx_high_local - 1;
w_high_cut = interp1(mag_meas_dB(idx_high-1:idx_high), ...
                     w_meas(idx_high-1:idx_high), cutoff_dB);

logw = log10(w_meas);

low_region = w_meas < (w_low_cut/3);
p_low = polyfit(logw(low_region), mag_meas_dB(low_region), 1);
slope_low = p_low(1);

high_region = w_meas > (w_high_cut*3);
p_high = polyfit(logw(high_region), mag_meas_dB(high_region), 1);
slope_high = p_high(1);

%% Estimated Transfer Function
hp_factor = sqrt(1/(sqrt(2)-1));
lp_factor = sqrt(sqrt(2)-1);

w1 = w_low_cut / hp_factor;
w2 = w_high_cut / lp_factor;

K_from_peak = 10^(peak_dB/20);
K = 10.0;

w = logspace(log10(min(w_meas)), log10(max(w_meas)), 4000).';
s = 1j*w;

H_est = K .* (s./(s+w1)).^2 .* (w2./(s+w2)).^2;
mag_est_dB = 20*log10(abs(H_est));
phase_est_deg = unwrap(angle(H_est))*180/pi;

H_est_meas = K .* ((1j*w_meas)./((1j*w_meas)+w1)).^2 .* ...
                 (w2./((1j*w_meas)+w2)).^2;
mag_est_meas_dB = 20*log10(abs(H_est_meas));
rms_error_dB = sqrt(mean((mag_meas_dB - mag_est_meas_dB).^2));

%% Load LTspice Exported TXT Data
% Uses imported MATLAB variables if they exist.
% Otherwise, it reads LTSpice_Output.txt directly.

if exist('Freq_', 'var') && exist('V_out__V_in_', 'var')
    lt_freq = Freq_;
    raw_lt = V_out__V_in_;
else
    rawFile = readcell('LTSpice_Output.txt', 'FileType', 'text');
    lt_freq = cell2mat(rawFile(2:end,1));
    raw_lt = string(rawFile(2:end,2));
end

raw_lt = string(raw_lt);
raw_lt = erase(raw_lt, ["(", ")", "'", "dB", "  "]);
parts_lt = split(raw_lt, ",");
```

```
lt_mag_dB = str2double(parts_lt(:,1));
lt_phase_deg = str2double(parts_lt(:,2));

lt_w = 2*pi*lt_freq;     % Hz to rad/s

% Optional unwrap for cleaner MATLAB phase plot
lt_phase_unwrapped_deg = unwrap(deg2rad(lt_phase_deg))*180/pi;

%% Print Results
fprintf('================ FINAL PROJECT RESULTS ================\n');
fprintf('Peak magnitude gain           = %.4f V/V\n', peak_mag_vv);
fprintf('Peak magnitude gain           = %.4f dB\n', peak_dB);
fprintf('Peak frequency                = %.4f rad/s\n', w_peak);
fprintf('-3 dB level                   = %.4f dB\n', cutoff_dB);
fprintf('Measured lower cutoff         = %.4f rad/s\n', w_low_cut);
fprintf('Measured upper cutoff         = %.4f rad/s\n', w_high_cut);
fprintf('Low-frequency slope           = %.4f dB/dec\n', slope_low);
fprintf('High-frequency slope          = %.4f dB/dec\n', slope_high);
fprintf('Estimated lower corner w1     = %.4f rad/s\n', w1);
fprintf('Estimated upper corner w2     = %.4f rad/s\n', w2);
fprintf('Gain from peak                = %.4f V/V\n', K_from_peak);
fprintf('Chosen design gain K          = %.4f V/V\n', K);
fprintf('RMS magnitude error           = %.4f dB\n', rms_error_dB);
fprintf('=======================================================\n\n');

fprintf('Estimated transfer function:\n');
fprintf('H(s) = %.6f * (s/(s + %.6f))^2 * (%.6f/(s + %.6f))^2\n\n', ...
    K, w1, w2, w2);

%%  Task 3: Raw Measured Data
fig1 = figure('Name','Task 3 - Raw Data','Color','w');
semilogx(w_meas, mag_meas_dB, 'b', 'LineWidth', 2);
grid on;
xlabel('Angular Frequency (rad/s)');
ylabel('Magnitude (dB)');
title('Measured Magnitude Response');
legend('Measured Data', 'Location', 'best');
set(gca,'FontSize',12);
saveas(fig1, 'Task3_RawData.png');

%%  Task 4: Picked Points
fig2 = figure('Name','Task 4 - Picked Points','Color','w');
semilogx(w_meas, mag_meas_dB, 'b', 'LineWidth', 2); hold on;
plot(w_peak, peak_dB, 'ro', 'MarkerSize', 10, 'LineWidth', 2);
plot(w_low_cut, cutoff_dB, 'ks', 'MarkerSize', 8, 'LineWidth', 2);
plot(w_high_cut, cutoff_dB, 'ks', 'MarkerSize', 8, 'LineWidth', 2);
yline(peak_dB, '--', sprintf('Peak = %.2f dB', peak_dB));
yline(cutoff_dB, '--g', sprintf('-3 dB = %.2f dB', cutoff_dB));
xline(w_low_cut, '--k', sprintf('\\omega_{cL}=%.3g rad/s', w_low_cut));
xline(w_high_cut, '--k', sprintf('\\omega_{cH}=%.3g rad/s', w_high_cut));
grid on;
xlabel('Angular Frequency (rad/s)');
ylabel('Magnitude (dB)');
title('Measured Data with Peak and Cutoff Frequencies');
legend('Measured Data','Peak','Cutoff Points','Location','best');
set(gca,'FontSize',12);
saveas(fig2, 'Task4_Picked_Points.png');

%%  Task 4: Estimated Magnitude
fig3 = figure('Name','Task 4 - Estimated Magnitude','Color','w');
semilogx(w, mag_est_dB, 'r', 'LineWidth', 2);
grid on;
xlabel('Angular Frequency (rad/s)');
ylabel('Magnitude (dB)');
title('Estimated Transfer Function Magnitude');
legend('Estimated Transfer Function', 'Location', 'best');
set(gca,'FontSize',12);
saveas(fig3, 'Task4_Estimated_Magnitude.png');

%% Task 4: Estimated Phase
fig4 = figure('Name','Task 4 - Estimated Phase','Color','w');
semilogx(w, phase_est_deg, 'm', 'LineWidth', 2);
grid on;
xlabel('Angular Frequency (rad/s)');
ylabel('Phase (deg)');
title('Estimated Transfer Function Phase');
legend('Estimated Phase', 'Location', 'best');
set(gca,'FontSize',12);
saveas(fig4, 'Task4_Estimated_Phase.png');

%%  Task 6: Magnitude Comparison
fig5 = figure('Name','Task 6 - Magnitude Comparison','Color','w');
semilogx(w_meas, mag_meas_dB, 'b', 'LineWidth', 2); hold on;
```

```
semilogx(w, mag_est_dB, 'r--', 'LineWidth', 2);
semilogx(lt_w, lt_mag_dB, 'k:', 'LineWidth', 2);
grid on;
xlabel('Angular Frequency (rad/s)');
ylabel('Magnitude (dB)');
title('Measured vs Estimated vs LTspice Magnitude Response');
legend('Measured Data','Estimated TF','LTspice Simulation','Location','best');
set(gca,'FontSize',12);
saveas(fig5, 'Task6_Magnitude_Comparison.png');

%% Task 7: Phase Comparison
fig6 = figure('Name','Task 7 - Phase Comparison','Color','w');
semilogx(w, phase_est_deg, 'm', 'LineWidth', 2); hold on;
semilogx(lt_w, lt_phase_unwrapped_deg, 'k--', 'LineWidth', 2);
grid on;
xlabel('Angular Frequency (rad/s)');
ylabel('Phase (deg)');
title('Estimated vs LTspice Phase Response');
legend('Estimated TF Phase','LTspice Phase','Location','best');
set(gca,'FontSize',12);
saveas(fig6, 'Task7_Phase_Comparison.png');

%% Save Results Text File
fid = fopen('automatic_results_final.txt','w');

fprintf(fid, '================================\n');
fprintf(fid, 'Peak magnitude gain          = %.6f V/V\n', peak_mag_vv);
fprintf(fid, 'Peak magnitude gain          = %.6f dB\n', peak_dB);
fprintf(fid, 'Peak frequency               = %.6f rad/s\n', w_peak);
fprintf(fid, '-3 dB level                  = %.6f dB\n', cutoff_dB);
fprintf(fid, 'Measured lower cutoff        = %.6f rad/s\n', w_low_cut);
fprintf(fid, 'Measured upper cutoff        = %.6f rad/s\n', w_high_cut);
fprintf(fid, 'Low-frequency slope          = %.6f dB/dec\n', slope_low);
fprintf(fid, 'High-frequency slope         = %.6f dB/dec\n', slope_high);
fprintf(fid, 'Estimated lower corner w1    = %.6f rad/s\n', w1);
fprintf(fid, 'Estimated upper corner w2    = %.6f rad/s\n', w2);
fprintf(fid, 'Gain from peak               = %.6f V/V\n', K_from_peak);
fprintf(fid, 'Chosen design gain K         = %.6f V/V\n', K);
fprintf(fid, 'RMS magnitude error          = %.6f dB\n', rms_error_dB);
fprintf(fid, '\nEstimated transfer function:\n');
fprintf(fid, 'H(s) = %.6f * (s/(s + %.6f))^2 * (%.6f/(s + %.6f))^2\n', ...
    K, w1, w2, w2);

fclose(fid);

disp('All plots saved automatically.');
disp('Results saved to automatic_results_final.txt');
```